

26 Reactor Design for Multiple Reactions

Most reactor design problems involve multiple reactions. Some reactions produce desired products, others produce undesired products, some reactions produce needed intermediates, some reactions deplete needed reactants/intermediates, some reactions produce harmful products, some reactions have significantly high activation energies but are highly exothermic and hence pose a problem with reaction runaway.

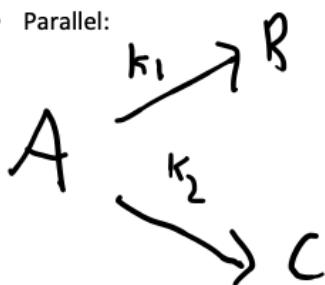
As chemical engineers, we need to concern ourselves with all reactions that *could* occur in a chemical reactor.

This includes reactions that are thermodynamically or kinetically suppressed under operating conditions, but which could come into play during circumstances where the reactor moves outside of the normal operating range.

In fact, most catastrophic reactor failures involve reactor designs that ignored such reactions.

Multiple reaction definitions:

- Parallel:



- assume elementary kinetics

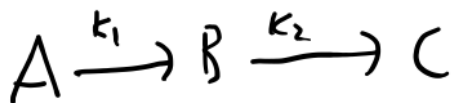
Rate laws:

$$-r_A = k_1 C_A + k_2 C_A = r_B + r_C \rightarrow \text{net rate of consumption of species A}$$

$$r_B = k_1 C_A$$

$$r_C = k_2 C_A$$

- Series:



- assume elementary

Rate laws:

$$-r_A = k_1 C_A$$

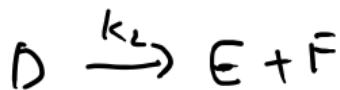
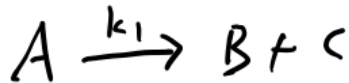
$$r_B = k_1 C_A - k_2 C_B$$

$$r_C = k_2 C_B$$

$$r_B = -r_A - r_C$$

"net rate of generation of species B"

• Independent:



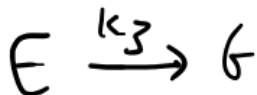
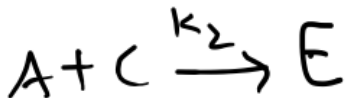
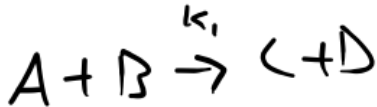
- assume elementary

Rate laws:

$$-r_A = k_1 C_A = r_B = r_C$$

$$-r_D = k_2 C_D = r_E = r_F$$

• Complex: combinations of serial, parallel, independent



- assume elementary

Rate laws:

Rate laws:

$$-r_A = k_1 C_A C_B + k_2 C_A C_C$$

$$-r_B = k_1 C_A C_B$$

$$r_C = k_1 C_A C_B - k_2 C_A C_C$$

$$r_D = k_1 C_A C_B$$

$$r_E = k_2 C_A C_C - k_3 C_E$$

$$r_G = k_3 C_E$$

- How does this change our procedure for solving reactor design problems?

- Constructing net rates → part of writing the rate law → to include multiple reactions

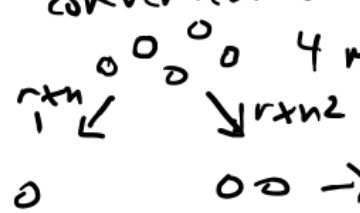
Need to include all relevant reactions and account for stoichiometry in the different reactions.

- Selecting the correct form of the mole balance

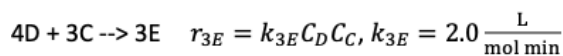
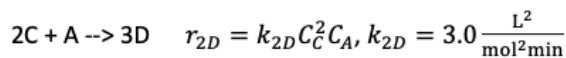
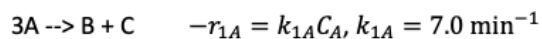
For the most part, you have to use N or F. The only exceptions that I can think of are:

- 1) A single reversible reaction, e.g., $A \rightleftharpoons B$
- 2) A single series of reactions, e.g., $A \rightarrow B \rightarrow C$.

The two reactions listed above can be solved in terms of X. All other problems involving multiple reactions have to use N or F. The reason for this is:

While X tells how much of a reactant has been converted in total, it doesn't tell how much was converted via the different reactions. E.g.  4 molecules of A initially
 $\rightarrow X_A = 0.75$

Let's do an example: The following reactions are taking place in liquid phase in a CSTR:



Write expressions for the net rates of A, B, C, D, and E, and use them to calculate the steady state rates in a CSTR that is operating at $C_A = 0.10 \text{ M}$, $C_B = 0.93 \text{ M}$, $C_C = 0.51 \text{ M}$, $C_D = 0.049 \text{ M}$.

$$r_A = r_{1A} + r_{2A} \rightarrow \frac{r_{2A}}{-1} = \frac{r_{2D}}{3} \rightarrow r_{2A} = -\frac{1}{3}r_{2D}$$

$$r_A = r_{1A} - \frac{1}{3}r_{2D}$$

$$r_B = -\frac{1}{3}r_{1A}$$

$$r_C = -\frac{1}{3}r_{1A} - \frac{2}{3}r_{2D} - r_{3E}$$

$$r_D = r_{2D} - \frac{4}{3}r_{3E}$$

$$r_E = r_{3E}$$

@ the given concentrations

$$r_A = -0.73 \frac{\text{mol}}{\text{L} \cdot \text{min}} \quad r_B = 0.233 \frac{\text{mol}}{\text{L} \cdot \text{min}} \quad r_C = 0.131 \frac{\text{mol}}{\text{L} \cdot \text{min}}$$

$$r_D = 0.0114 \frac{\text{mol}}{\text{L} \cdot \text{min}} \quad r_E = 0.05 \frac{\text{mol}}{\text{L} \cdot \text{min}}$$

Let's take this same example. If the volumetric flowrate is 100 L/min and the entering concentration of A is 3 M, what is the volume of the CSTR?

mole balances:

$$0 = F_{i0} - F_i + V r_j$$

$$F_{A0} = C_{A0} v_0$$

$$F_i = C_j v_0$$